

Dear Hiring Manager,

I'm a Bachelor of Science in Computer Science student at Columbia University in the City of New York. I have experience across financial services, web development, software development, and AI research, and I'm particularly motivated by opportunities to build scalable, reliable infrastructure that directly improves product performance.

In my current role as a Web Developer Intern with Columbia College Information Technology, I've been responsible for migrating 43 departmental websites from Drupal 7 to Drupal 10. That work has involved writing migration scripts, refactoring custom modules, conducting regression testing, and optimizing performance. As a result, we improved average page load times by about 35% and eliminated all deprecated modules in production. I also integrated a Django/PostgreSQL/REST backend with a new Columbia admissions website, scaling to 50,000+ applicants per cycle and reducing form submission errors by roughly 40%, which significantly cut manual data correction effort.

Beyond web infrastructure, I have strong experience in applied AI. At the Columbia University Artificial Intelligence Laboratory, I led the development of gAyl, an AI tool for LGBTQ+ suicide prevention. I built LLM-based classifiers in PyTorch and Hugging Face. Our system achieved a 15% false-negative rate on unsafe-response detection across more than 1,000 high-risk prompts, and the work culminated in a research publication.

As a remote Software Developer Intern at Zitouna Bank, I engineered RESTful APIs using Python and Django and collaborated closely with database teams to tune PostgreSQL for high-volume financial data, which resulted in API response times being cut by 30%, increased transaction throughput by 25%, and reduced p95 query latency by about 35%, while maintaining a 100% compliance rate during security and regulatory audits.

Since 2021, I've also worked as a freelance developer, helping more than 50 clients customize web applications, integrate APIs, optimize performance, and implement cybersecurity measures. This freelance work sharpened my ability to quickly understand requirements, communicate clearly with non-technical stakeholders, and deliver reliable solutions on tight timelines.

I would sincerely appreciate the opportunity to further discuss my background and skills. Thank you for your time and consideration.

Sincerely,
Sammy Abdella